

Operating Systems

BCA — Semester IV — 2023 — Answer Sheet

Subject Code: CACS251

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Define Operating System? Explain objectives of Operating System.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How can we solve producer-consumer problem using semaphore?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What do you mean by Preemptable and Non-Preemptable resources? Describe the necessary conditions for deadlock.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Given memory partitions of 200k, 600k, 300k, 400k, and 700k (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 315k, 527k, 202k, and 548k (in order)? Which algorithm makes the most efficient use of memory?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is meant by system threat? Explain different types of program threats.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain methods of communication in a distributed system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on (Any Two):a) Access Control Matrixb) Process Statesc) Kernel

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. The processes are assumed to have arrived in the order p1, p2, p3, p4, p5 all at time 0. Draw a Gantt chart illustrating their execution and calculate average waiting time and turnaround time for the process given below using:
a) First Come First Serve
b) Shortest Remaining Time Next
c) Priority Scheduling
d) Round Robin (quantum = 2)
Processes

Process	Burst Time (sec)	Priority
P1	22	2
P2	11	1
P3	8	4
P4	4	2
P5	3	5

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is Disk Access Time and Disk Response Time? Suppose that a disk has 200 cylinders, numbered (0-199). The drive is currently serving a request at cylinder 43/and previous request was at cylinder 25. The queue of pending request, in FIFO order is: 55, 58, 39, 18, 90, 160, 150, 38, 184. Starting from the current head position, what is total distance (in cylinders) that the disk arm moves to satisfy all pending request for each of following disk scheduling algorithms?
a) FCFS
b) SSTF
c) C-SCAN
d) LOOK

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Differentiate between internal and external fragmentation? Explain briefly with examples and procedure to remove the types of fragmentation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.